

PGE AND Au IN ORE-FREE DOLERITE SILLS OF THE SIBERIAN TRAPS: ESTIMATION OF INITIAL METAL CONCENTRATIONS IN MAFIC MELTS

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The classic example of sulfide Cu-Ni±Pt-Pd deposits located within mafic intrusions is the Noril'sk-Talnakh group of deposits in the northwestern Siberian Platform. It is commonly assumed that sulfide ores are formed by liquation (Ryabov et al., 2014). Many sulfides of such deposits, however, are characterized by heavy sulfur isotopes (Grinenko, 1985; Li et al., 2003). This fact is interpreted in favor of contamination of mafic melts by sulfur from sulfate-rich sediments and sulfur-rich gases (Grinenko, 1985; Li et al., 2003).

The source of metals in sulfide melts is questionable. Again, it is commonly assumed that metals are derived from mantle and when they meet crustal sulfur, a sulfide melt is formed within silicate melt matrix. Heavy sulfide melt sinks to the bottom of a magmatic conduit, while remained barren silicate melt flows away. We wish to test this model for the Noril'sk-Talnakh group of deposits by the mass balance calculations in the following way; (1) estimate the metals' concentrations in initial mafic melts, (2) estimate the overall metals' abundance within the deposits, (3) calculate the volume of mafic melts required to produce the deposits by knowing the first two parameters, (4) compare the calculated melt volume with the available volume of the intrusive and extrusive phases in the region of interest. Each of these four steps is a nontrivial task. Thus, this study aims to the first step only.

What rocks should be taken for analysis to estimate concentrations of PGE and Au in initial melts? Intrusions in the Noril'sk-Talnakh area are rich in these metals, whereas barren intrusions, according to the explained model above, may be existed by the elements of interest. It was shown, that intrusions in the Angara-Taseeva syncline of the southern Siberian Traps province, on one hand, free of ore component and, on the other hand, have geochemical analogues among extrusive formations of the Noril'sk-Talnakh area (Ivanov et al., 2008). So, they were taken as the case study.

We collected 17 samples of dolerite sills with large MgO variations from surface outcrops within the Angara-Taseeva syncline. These samples were analyzed for major, trace elements, including PGE and Au, and Sr-Nd isotopes. According to chemistry and Sr-Nd isotopes, the dolerite sills of the Angara-Taseeva syncline resemble low-Ti basalts of the Noril'sk-Talnakh area and, except PGE and Au concentrations, do not differ from ore-bearing and barren intrusions of the same region.

Table 1. Estimated concentrations of PGE and Au (ppb) for mafic melts.

	High MgO	Most differentiated	Average
Au	3	12	4
Pt	5-10	33	11
Pd	9-10	40	14
Rh	n.e.	n.e.	0.6
Ru	n.e.	n.e.	0.2
Ir	n.e.	n.e.	0.06

n.e. – not estimated, because there is no significant variations among different compositions

As for PGE and Au, there are no significant variations for Ir, Ru, Rh among different by chemistry dolerite sills. On average, concentrations of these elements are at sub-ppb level (Table 1). Au correlates positively with TiO₂ and FeO_{total} (Fig. 1) and negatively with MgO and Mg#. This shows that in sulfur-undersaturated melts Au behaves as incompatible element and accumulates in course of differentiation. The difference between the most differentiated and the most primitive rock samples is

fourfold (Table 1). The difference in TiO_2 and FeO_t concentrations is ~ 5 -fold and ~ 2 -fold, suggesting that the major crystallizing mineral phases were olivine, plagioclase and orthopyroxene in agreement with petrographic and geochemical observations (Feoktistov, 1978; Ivanov et al., 2008).

Pt and Pd behave similarly to Au. In general, their concentrations also increase, though some sample deviate from this behavior (Fig. 1). Probably in course of differentiation, sulfur concentrations also increases and if a threshold value is reached it immediately form Pt or Pd sulfide phase.

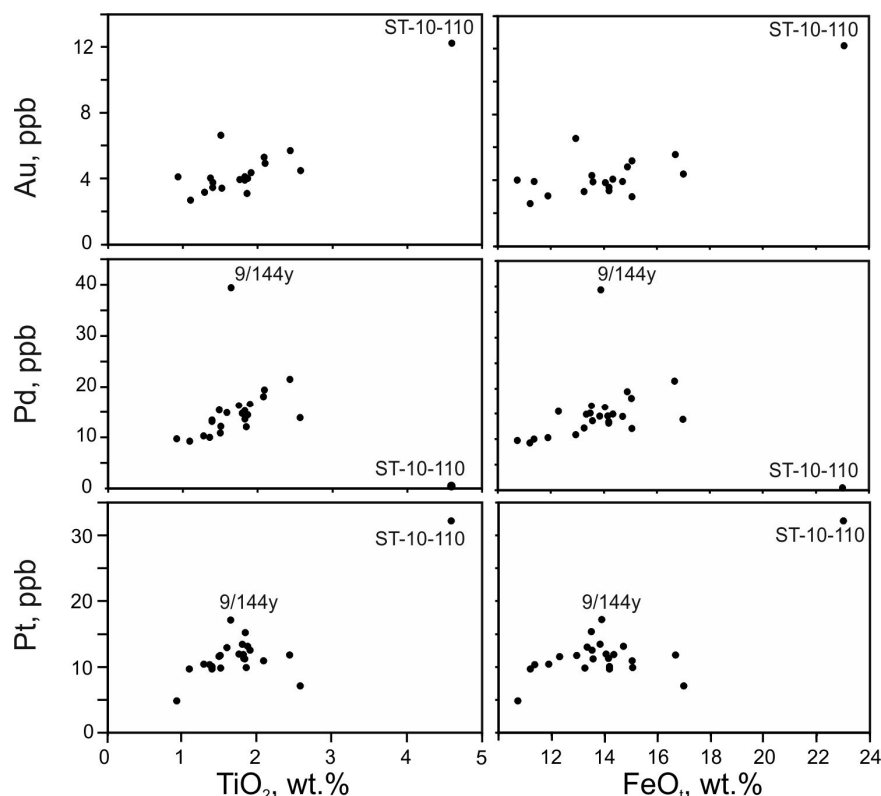


Figure 1. Variations of Pt, Pd and Au vs TiO_2 and FeO_t in dolerite sills of the Angara-Taseeva syncline, southeastern Siberian Traps province. Sample numbers of some extreme sample are shown.

Considering the trends in Fig. 1, we estimate probable concentrations of Pt, Pd and Au for the most primitive and the most differentiated rocks (Table 1). Generally speaking, the most primitive rocks are characterized by 3 ppb Au, 5-10 ppb Pt and 9-10 ppb Pd. These should be used as initial concentrations for any further consideration of ore process in the Noril'sk-Talnakh ore district in regards the testing liquation model.

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